

ALEXANDER MATUS, Ph.D.

SENIOR GEOSPATIAL DATA ENGINEER

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SUMMARY

Remote Sensing Data Engineer and Atmospheric Scientist with 15+ years of experience architecting scalable pipelines for NASA's most complex geospatial datasets. Expert in LIDAR, petabyte-scale data ingestion, and multi-sensor data fusion. Proven track record of translating high-dimensional observations into production-level AI/ML algorithms for environmental risk monitoring, wildfire smoke tracking, and resource management.

TECHNICAL SKILLS

Programming: Python (Expert: xarray, Dask, Pandas), MATLAB, C++, Fortran, SQL, Github
Machine Learning: PyTorch, TensorFlow/Keras, Computer Vision, Image Classification, Uncertainty Analysis
Geospatial: PostGIS, GDAL/OGR, Rasterio, Geopandas, Google Earth Engine, GeoTIFF
Data Engineering: Scalable ETL Pipeline Design, Distributed Computing (Dask/AWS), HDF5/NetCDF
Domain Expertise: LIDAR, Imagery, Wildfire Aerosol Tracking, Radiative Transfer, Multi-sensor Fusion

EXPERIENCE

Assistant Research Scientist

2023 - present

NASA Goddard Space Flight Center

Greenbelt, MD

- **Geospatial Infrastructure:** Architected high-performance Python/Dask pipelines to ingest and analyze petabyte-scale datasets from multiple satellite constellations (MODIS, CERES, GPCP).
- **Product Development:** Developed automated workflows to quantify global atmospheric energy budgets, directly informing climate resilience and Earth system models.
- **Operational Leadership:** Served as Instrument Scientist for major airborne campaigns (IMPACTS, ACCLIP); managed real-time sensor calibration and data integrity for high-stakes stakeholders.

NASA Postdoctoral Fellow

2021 - 2023

NASA Goddard Space Flight Center

Greenbelt, MD

- **ML for Risk Detection:** Implemented deep learning frameworks (TensorFlow/Keras) to classify and predict environmental risks by analyzing 3D LIDAR point clouds and vertical profiles.
- **Wildfire/Aerosol Analysis:** Engineered a novel retrieval method for tracking long-range aerosol transport (including wildfire smoke), securing \$170k in competitive funding.
- **Sensor Fusion:** Developed multi-instrument data fusion algorithms combining active LIDAR with passive multispectral imagery to enhance feature detection in complex environments.

CloudSat Algorithm Developer

2016 - 2021

University of Wisconsin-Madison/CIMSS

Madison, WI

- **Production Algorithms:** Led development of version upgrades and operational deployment of the CloudSat FLXHR-LIDAR, a core NASA data product used globally by the research community.
- **Analytic Dashboards:** Designed real-time automated cloud classification software and interactive Plotly/Dash web applications for 3D satellite data visualization.
- **Validation Frameworks:** Conducted rigorous uncertainty quantification for cloud and aerosol radiative forcing estimates, resulting in 30+ peer-reviewed publications.

Research Assistant (GIS Specialist)
NASA Langley Research Center (DEVELOP/LARSS)

2008 - 2009
Hampton, VA

- **Resource Management:** Developed GIS and cartographic products for the National Park Service and National Geographic to prioritize conservation and resource allocation.
- **Hazard Detection:** Performed analysis of volcanic ash detection and long-range transport for applied science study on economic impacts affecting the commercial aviation industry.

EDUCATION

Ph.D., Atmospheric and Oceanic Sciences, University of Wisconsin-Madison
M.S., Atmospheric and Oceanic Sciences, University of Wisconsin-Madison
B.S., Meteorology (honors) and GIS, Pennsylvania State University

2016
2013
2009

OPERATIONAL ROLES

Instrument Engineer
NASA Goddard Space Flight Center

2022 - 2023
Greenbelt, MD

- Managed real-time lidar instrumentation on research aircraft for IMPACTS (Investigation of Microphysics and Precipitation for Atlantic Coast-Threatening Snowstorms) campaign in Atlanta, GA.

Instrument Engineer
NASA Goddard Space Flight Center

2021 - 2022
Greenbelt, MD

- Managed real-time lidar calibration and sensor response in support of ACCLIP (Asian Summer Monsoon Chemical & Climate Impact Project) aircraft field campaign at Osan Air Base, South Korea.

Algorithm Lead (CloudSat)
University of Wisconsin-Madison/CIMSS

2016 - 2021
Madison, WI

- Directed version upgrades and operational implementation of CloudSat radiative fluxes products (2B-FLXHR-LIDAR) at the UW-Madison and Cooperative Institute for Meteorological Satellite Studies.

SELECTED PUBLICATIONS

Matus, A., R. Kramer, L. Oreopoulos, and N. Cho (2026). Improving Climatological Atmospheric Energy Budget Closure Amidst Interannual Precipitation Volatility. *Geophysical Research Letters*. (submitted).

Matus, A., Nowottnick, E. P., Yorks, J., and da Silva, A. (2025). Enhancing surface PM_{2.5} air quality estimates in GEOS using CATS lidar data. *Earth and Space Science*, 12, e2024EA004078. doi: 10.1029/2024EA004078.

Matus, A. and T. L’Ecuyer (2017). “The role of cloud phase in Earth’s radiation budget.” *Journal of Geophysical Research: Atmospheres*, 122, pp. 2559–2578, doi: 10.1002/2016JD025951.

Total Citations: 700+ | **Publications:** 35

AWARDS & HONORS

- **Group Achievement Award IMPACTS Team,** NASA Goddard Space Flight Center (2025).
- **NASA Postdoctoral Fellowship,** NASA Goddard Space Flight Center (2021).
- **Lettau Award for Outstanding PhD Thesis,** UW-Madison (2017).